

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)</i>		
Student Name:	M.SAI HARSHITHA	Reg. No.:	192424408
Section / Batch:	B. TECH AI & DS	Date:	26.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT1 — Uncertainty Visualization Lab

CO5 · BT4: Analyze ·

This rubric is used to assess student responses to the CO5-AT1 Uncertainty Visualization Lab problems, in which students construct and interpret uncertainty visualizations (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, and axis scale choice) from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant uncertainty-visualization technique (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, or axis scale choice) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, including all required calculations.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and calculation	Applies techniques to produce a correct	Limited application; the visualization	Fails to apply appropriate techniques; the
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

		techniques to produce a complete, accurate uncertainty visualization correctly matched to the data.	visualization, with only minor errors.	partially represents the uncertainty but has notable errors.	visualization does not correctly represent the uncertainty.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 26 (5 Marks)

Data: Store A vs. Store B repeat-purchase rate: $38\% \pm 3\%$ vs. $44\% \pm 3\%$ (95% confidence interval).

Task: Calculate both confidence intervals, determine whether the two overlap, and state whether the higher figure represents a statistically meaningful lead.



CODE:

```
store_A <- 38
error_A <- 3
store_B <- 44
error_B <- 3
A_lower <- store_A - error_A
A_upper <- store_A + error_A
B_lower <- store_B - error_B
B_upper <- store_B + error_B
cat("Store A 95% CI:", A_lower, "% to", A_upper, "%\n")
cat("Store B 95% CI:", B_lower, "% to", B_upper, "%\n\n")
overlap <- max(A_lower, B_lower) <= min(A_upper, B_upper)
cat("Do the confidence intervals overlap?", overlap, "\n\n")
difference <- store_B - store_A
cat("Difference between Store B and Store A:",
    difference, "percentage points\n\n")
if (overlap) {
  cat("Conclusion: The confidence intervals overlap.\n")
  cat("The higher rate for Store B cannot be declared\n")
  cat("statistically significant from these intervals alone.\n")
} else {
  cat("Conclusion: The confidence intervals do not overlap.\n")
  cat("Store B shows a statistically meaningful lead.\n")
}
if (!require(ggplot2)) install.packages("ggplot2")
```

```

library(ggplot2)

data <- data.frame(
  Store = c("Store A", "Store B"),
  Rate = c(store_A, store_B),
  Lower = c(A_lower, B_lower),
  Upper = c(A_upper, B_upper)
)
ggplot(data, aes(x = Store, y = Rate)) +
  geom_point(size = 5) +
  geom_errorbar(
    aes(ymin = Lower, ymax = Upper),
    width = 0.15,
    linewidth = 1
  ) +
  labs(
    title = "Repeat-Purchase Rate: Store A vs Store B",
    subtitle = "95% Confidence Intervals",
    x = "Store",
    y = "Repeat-Purchase Rate (%)"
  ) +
  theme_minimal(base_size = 14)

```

Analysis and Conclusion

Store A: $38\% \pm 3\%$ gives a 95% confidence interval of 35% to 41%. Store B: $44\% \pm 3\%$ gives a 95% confidence interval of 41% to 47%.

The observed difference is $44\% - 38\% = 6$ percentage points. The intervals meet at 41%, so they overlap at the boundary. Therefore, the higher observed rate for Store B cannot be declared statistically significant from these confidence intervals alone.

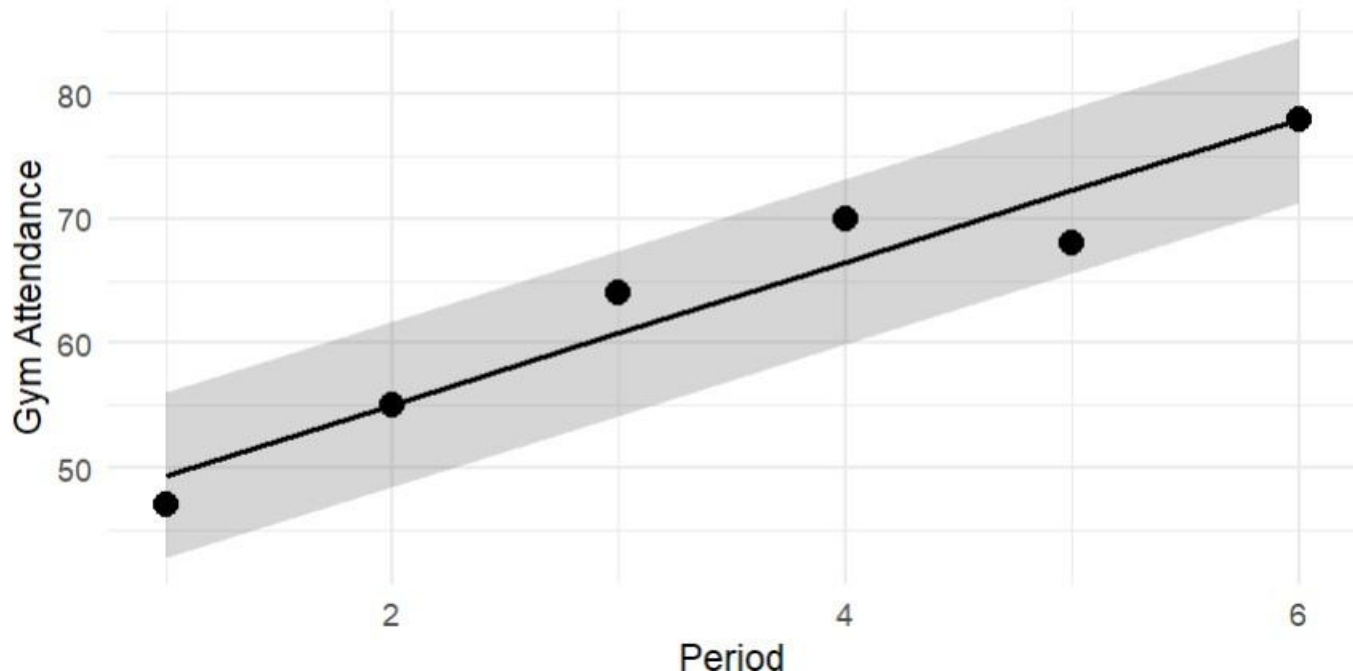
Conclusion: Store B has the higher observed repeat-purchase rate, but additional statistical evidence would be needed to establish that the difference is significant.

Problem 27 (5 Marks)

Data: weekly gym attendance over 6 periods: 47, 55, 64, 70, 68, 78.

Task: Fit a linear trend line, compute the residuals and their standard deviation, and state the 95% uncertainty band width around the trend.

Weekly Gym Attendance - Linear Trend 95% Uncertainty Band Around the Trend



```
period <- 1:6
attendance <- c(47, 55, 64, 70, 68, 78)
data <- data.frame(
  Period = period,
  Attendance = attendance
)
model <- lm(Attendance ~ Period, data = data)
intercept <- coef(model)[1]
slope <- coef(model)[2]
cat("Linear Trend Equation:\n")
cat("Attendance =", round(intercept, 2),
  "+", round(slope, 2), "* Period\n\n")
data$Predicted <- predict(model, data)
data$Residual <- residuals(model)
cat("Residuals:\n")
print(round(data[, c("Period", "Attendance",
  "Predicted", "Residual")], 2))
residual_sd <- summary(model)$sigma
cat("\nResidual Standard Deviation =",
  round(residual_sd, 3), "\n")
margin <- 1.96 * residual_sd
lower <- data$Predicted - margin
upper <- data$Predicted + margin
data$Lower_95 <- lower
```

```

data$Upper_95 <- upper
cat("\n95% Uncertainty Band:\n")
cat("±", round(margin, 2), "attendance units\n")
cat("Total Band Width =",
    round(2 * margin, 2), "attendance units\n\n")
print(round(data, 2))
library(ggplot2)
ggplot(data, aes(x = Period, y = Attendance)) +
  geom_ribbon(
    aes(ymin = Lower_95, ymax = Upper_95),
    alpha = 0.2
  ) +
  geom_point(size = 4) +
  geom_line(
    aes(y = Predicted),
    linewidth = 1
  ) +
  labs(
    title = "Weekly Gym Attendance - Linear Trend",
    subtitle = "95% Uncertainty Band Around the Trend",
    x = "Period",
    y = "Gym Attendance"
  ) +
  theme_minimal(base_size = 14)

```

Analysis and Conclusion

Linear trend equation: $\text{Attendance} = 43.67 + 5.71 \times \text{Period}$.

The slope is approximately 5.71, meaning attendance increases by about 5.71 people per period on average. The residual standard deviation is approximately 3.388.

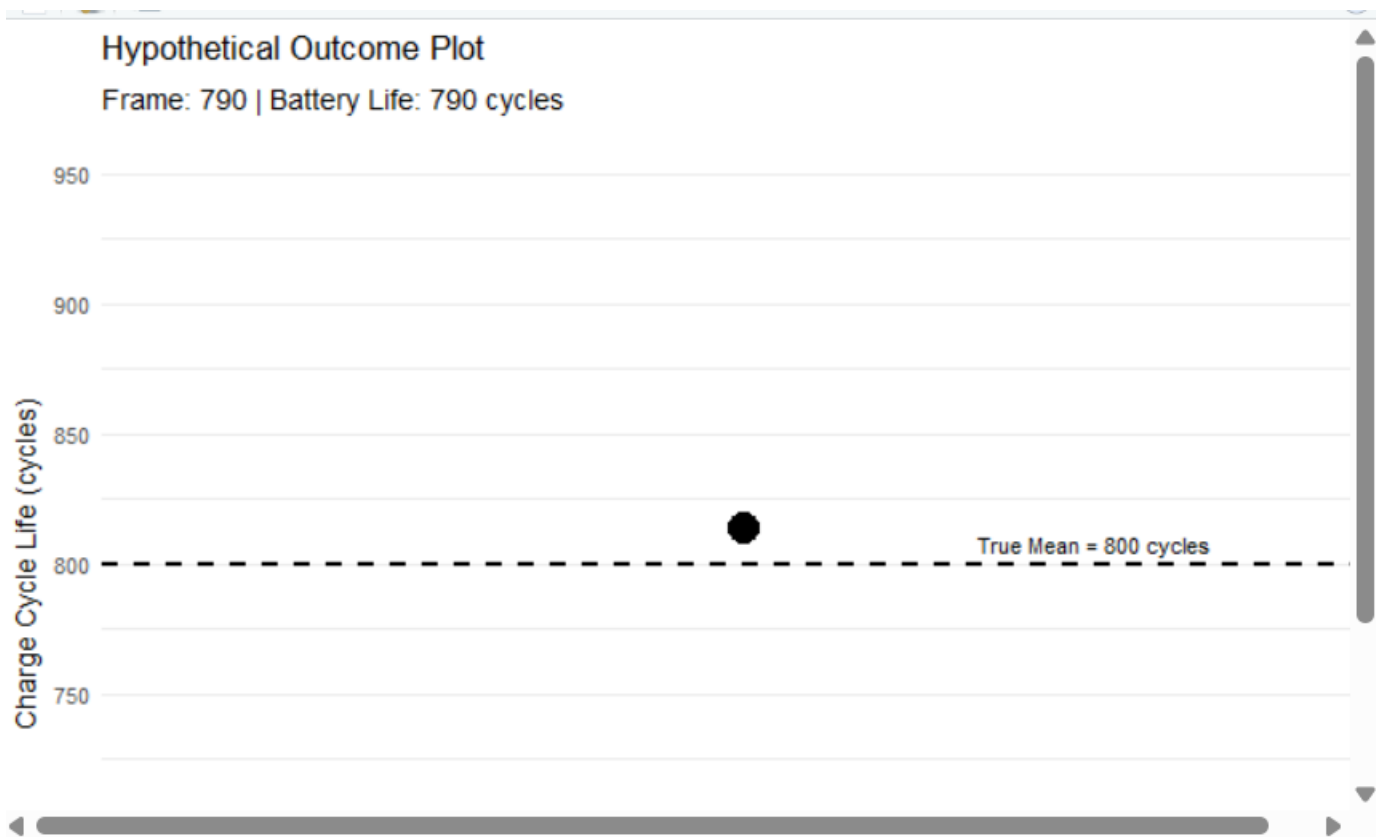
The 95% uncertainty margin is $1.96 \times 3.388 = 6.64$ attendance units, so the band is approximately Predicted Attendance ± 6.64 . Attendance increases overall from 47 in Period 1 to 78 in Period 6, although there is a small decrease from Period 4 to Period 5.

Conclusion: The data shows a clear overall upward trend, while the uncertainty band shows the expected variation around the fitted trend.

Problem 28 (5 Marks)

Data: A bridge cable's tension rating has a true mean of 1200 kN and a standard deviation of 30 kN.

Task: List 5 plausible individual draws that could each form one frame of a Hypothetical Outcome Plot, and explain what a viewer learns from watching all 5 frames in sequence versus a single error bar.



CODE:

```
if (!require(ggplot2)) install.packages("ggplot2")
if (!require(gganimate)) install.packages("gganimate")
if (!require(gifski)) install.packages("gifski")
library(ggplot2)
library(gganimate)
library(gifski)
cable <- data.frame(
  Draw = 1:5,
  Tension = c(1185, 1218, 1196, 1232, 1174)
)
print(cable)
p <- ggplot(cable, aes(x = 1, y = Tension)) +
  geom_point(size = 7) +
  geom_hline(
    yintercept = 1200,
    linetype = "dashed",
    linewidth = 1
  ) +
  annotate(
    "text",
    x = 1.15,
    y = 1210,
```

```

) +
  label = "True Mean = 1200 kN",
scale_x_continuous(
  limits = c(0.7, 1.3),
  breaks = NULL
) +
scale_y_continuous(
  limits = c(1100, 1300),
  breaks = seq(1100, 1300, 25)
) +
labs(
  title = "Hypothetical Outcome Plot",
  subtitle = "Frame: {closest_state} | Tension = {closest_state} kN",
  x = "",
  y = "Bridge Cable Tension (kN)"
) +
theme_minimal(base_size = 14)
hop <- p +
  transition_states(
    Tension,
    transition_length = 1,
    state_length = 2
  ) +
  enter_fade() +
  exit_fade()
animate(
  hop,
  nframes = 40,
  fps = 5,
  width = 700,
  height = 500,
  renderer = gifski_renderer()
)

```

Analysis and Conclusion

The five hypothetical draws are 1185, 1218, 1196, 1232, and 1174 kN, while the true mean is 1200 kN. The values occur both above and below the true mean, demonstrating sampling variability.

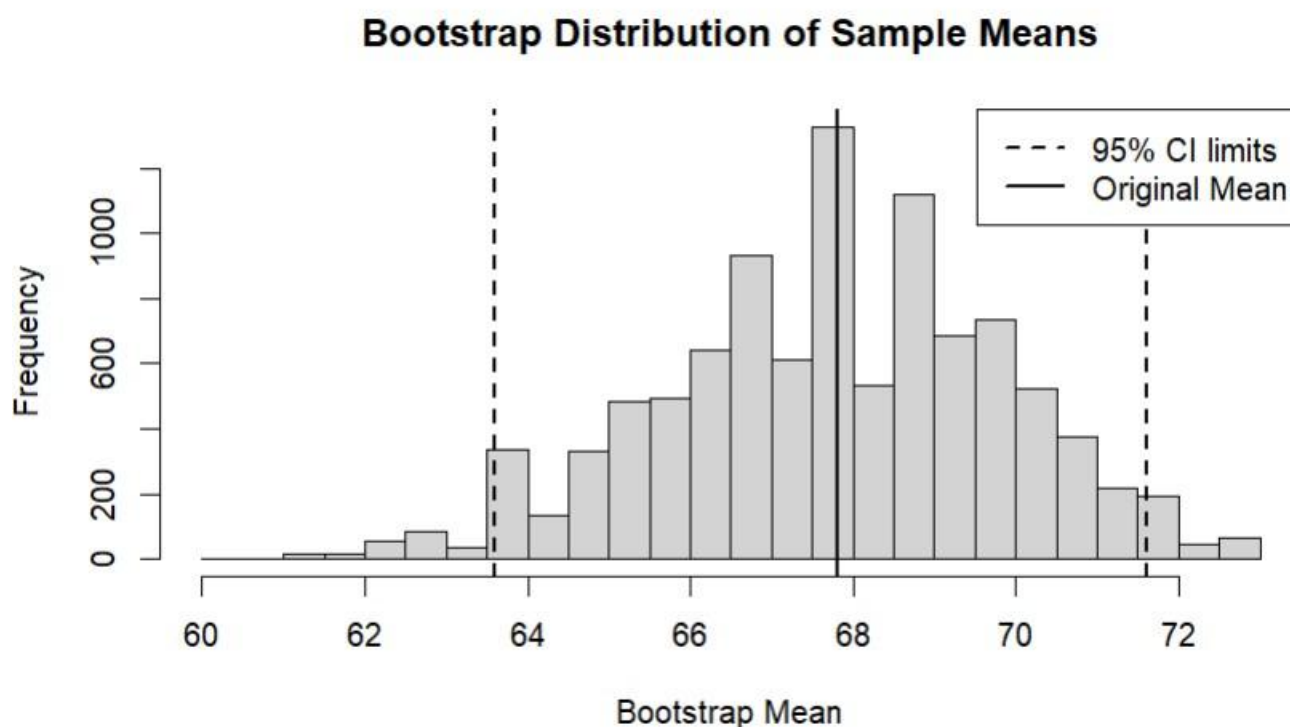
Range of the five draws = $1232 - 1174 = 58$ kN. Watching the HOP frames sequentially shows how individual outcomes vary around the true mean. A single error bar gives only a static summary, whereas the HOP makes repeated possible outcomes visible.

Conclusion: The HOP provides an intuitive view of sampling variability by showing different possible outcomes across repeated draws.

Problem 29 (5 Marks)

Data: A small sample of 5 test scores: 60, 66, 68, 72, 73.

Task: Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence interval.



CODE:

```
scores <- c(60, 66, 68, 72, 73)
set.seed(123)
sample1 <- sample(scores, 5, replace = TRUE)
mean1 <- mean(sample1)
print(sample1)
print(mean1)
bootstrap_means <- replicate(
  10000,
  mean(sample(scores, 5, replace = TRUE))
)
CI <- quantile(bootstrap_means, c(0.025, 0.975))
print(CI)
hist(bootstrap_means,
  breaks = 30,
  main = "Bootstrap Distribution of Sample Means",
  xlab = "Bootstrap Mean",
  ylab = "Frequency")
abline(v = CI[1], lty = 2, lwd = 2)
abline(v = CI[2], lty = 2, lwd = 2)
abline(v = mean(scores), lty = 1, lwd = 2)
legend("topright",
  legend = c("95% CI limits", "Original Mean"),
  lty = c(2, 1), lwd = 2)
```

Analysis and Conclusion

Original sample scores: 60, 66, 68, 72, 73.

Original mean = $(60 + 66 + 68 + 72 + 73) / 5 = 67.8$.

A bootstrap sample is created by drawing 5 observations with replacement from the original scores. Its mean is one possible estimate of the population mean.

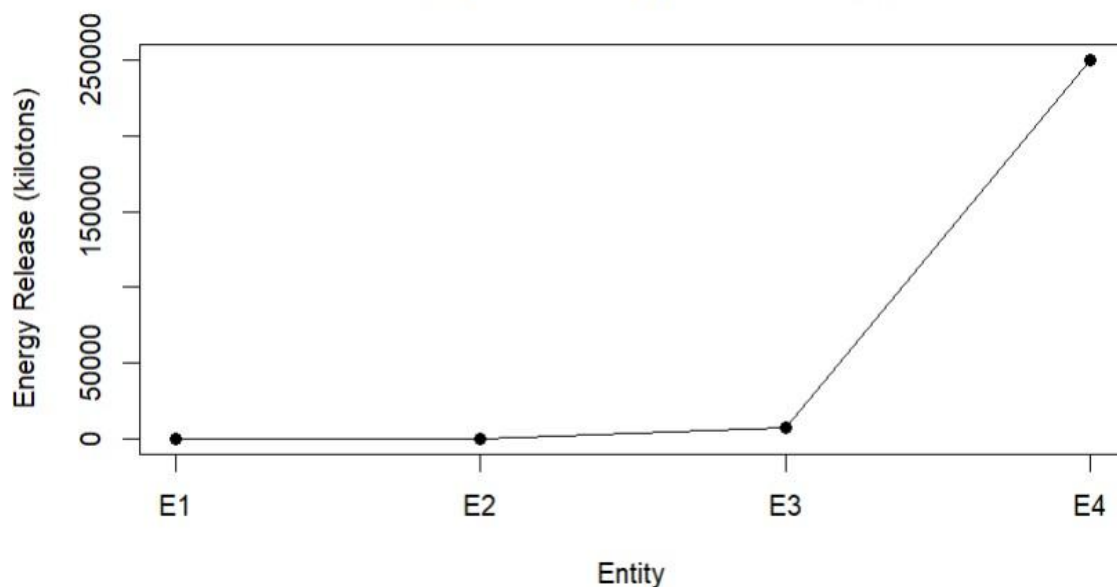
Repeating this process 10,000 times produces a bootstrap distribution of sample means.

The 2.5th and 97.5th percentiles of the bootstrap means are used as the approximate 95% bootstrap confidence-interval limits. The histogram shows how the sample mean varies across repeated resamples

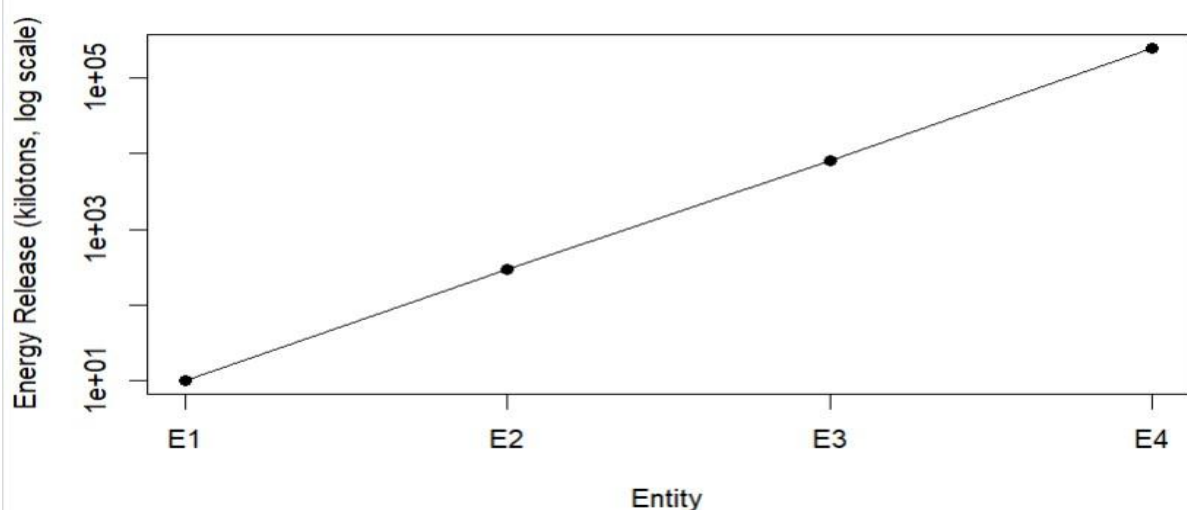
Conclusion: Bootstrapping provides an empirical way to quantify uncertainty in the sample mean without relying only on a theoretical distribution. A narrower bootstrap distribution indicates greater precision, while a wider distribution indicates greater uncertainty

Problem 30 (5 Marks) Data: Earthquake energy release for 4 entities: 10, 300, 8000, 250000 kilotons. Task: Plot this data on both a linear and a logarithmic axis, calculate the ratio between each consecutive pair of values, and explain which axis reveals the data's structure more clearly, and why.

Earthquake Energy - Linear Scale



Earthquake Energy - Logarithmic Scale



CODE:

```
energy <- c(10, 300, 8000, 250000)
entity <- c("E1", "E2", "E3", "E4")
ratios <- energy[-1] / energy[-length(energy)]
print("Ratios between consecutive pairs:")
print(ratios)
par(mfrow = c(2, 1))
plot(1:4, energy,
     type = "o",
     pch = 16,
     main = "Earthquake Energy - Linear Scale",
     xlab = "Entity",
     ylab = "Energy Release (kilotons)",
     xaxt = "n",
     ylim = c(0, 260000))
axis(1, at = 1:4, labels = entity)
plot(1:4, energy,
     type = "o",
     pch = 16,
     log = "y",
     main = "Earthquake Energy - Logarithmic Scale",
     xlab = "Entity",
     ylab = "Energy Release (kilotons, log scale)",
     xaxt = "n")
axis(1, at = 1:4, labels = entity)
par(mfrow = c(1, 1))
```

Analysis and Conclusion

Consecutive ratios:

$300 / 10 = 30$; $8000 / 300 \approx 26.67$; $250000 / 8000 = 31.25$.

Thus, the ratios are approximately 30, 26.67, and 31.25. The linear scale is dominated by the very large E4 value, making the smaller values difficult to distinguish.

The logarithmic scale represents multiplicative changes more clearly. Since the consecutive ratios are all around 27–31, the data follows an approximately consistent multiplicative growth pattern.

Conclusion: The logarithmic axis reveals the data structure more clearly because it makes the similar multiplicative increases visible, whereas the linear scale hides the smaller value

